LEARNER GUIDE

What is a digit worth?

Place Value · Lesson 1. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

A **digit** is one of the ten symbols 0 to 9. What a digit is worth depends on the **column** it stands in. In 4,073,250 the 7 is in the ten-thousands column, so it is worth 70,000. Every column is worth **ten times** the column to its right, because ten of one column make one of the next. Large numbers are easier to read in groups of three: millions, thousands, ones.

What you are learning to notice

- The digit stays the same; only its column changes what it is worth.
- One column to the left: ten times as much. One column to the right: a tenth as much.
- A 0 is worth nothing, but it holds its column open. Without it, 4,050 would become 450.
- Split a large number into groups of three from the right to read it: 2,040,500 is two million, forty thousand, five hundred.
- To make the largest number from some digits, put the largest digit in the largest column.

Moving through it, one step at a time

To find what a digit is worth, write the number in a place-value chart, one digit per column. Read the heading above the digit: the digit times that column's value is what it is worth. To read a large number, split it into groups of three from the right, and read each group followed by its name: million, thousand, then the rest.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — What is each digit worth?

In 4,073,250, what is the 7 worth? And what is the 0 after the 4 doing?

Millions	Hundred thousands	Ten thousands	Thousands	Hundreds	Tens	Ones
4	0	7	3	2	5	0

The 7 is in the ten-thousands column.

1. Write the number in the chart, one digit per column.	4 | 0 | 7 | 3 | 2 | 5 | 0
2. Read the heading above the 7.	ten thousands
3. Multiply the digit by its column.	7 × 10,000 = 70,000
4. Look at the 0 after the 4.	it holds the hundred-thousands column open

The 7 is worth 70,000, seventy thousand. Without the 0, the number would be 473,250.

Example 2 — Sliding a digit

What happens to the value of a 7 when it moves one column to the left?

7 in the hundreds column

Millions	Hundred thousands	Ten thousands	Thousands	Hundreds	Tens	Ones
				7	0	0

Slide it one column left

Millions	Hundred thousands	Ten thousands	Thousands	Hundreds	Tens	Ones
			7	0	0	0

1. In the hundreds column it is worth	700
2. One column left, in the thousands column	7,000
3. Compare the two values.	7,000 = 700 × 10
4. Why?	a thousand is ten hundreds

Each step to the left makes a digit worth ten times as much, because each column holds ten of the column to its right.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (place value up to the thousands), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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